

VENKAT AKHIL ANKEM

Operations Research · Large-Scale Optimization · Applied Machine Learning

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PROFESSIONAL SUMMARY

Ph.D. candidate in Industrial Engineering (Polytechnique Montréal / GERAD) with experience spanning operations research, applied machine learning, and industrial analytics. MITACS-funded industrial collaborator with Rio Tinto — delivered cutting-plane and rolling-horizon methods achieving 90% reduction in mine-scheduling computation time and 80% reduction in time-to-feasibility, deployed directly into Rio Tinto planning workflows. Published in Computers & Operations Research (2026); a second paper is under review at Engineering Optimization; 5 peer-reviewed journal publications in total. Currently developing machine learning models to predict MILP instance difficulty and guide solver strategy on industrial-scale problems (100,000+ variables), and maintains a public portfolio of applied ML/RL projects. Seeking Optimization Scientist, ML Optimization Engineer, or Decision Support roles where rigorous mathematical methods and applied ML drive operational impact.

CORE COMPETENCIES

Mathematical Optimization	Algorithms & Heuristics	Applied ML & Software
Mixed-Integer Programming (MILP/MIP) Cutting Planes & Model Strengthening LP Relaxations, Branch-and-Cut Combinatorial Optimization	Large Neighborhood Search (LNS) Relax-and-Repair Frameworks Rolling-Horizon Decomposition ML-Guided Optimization	PyTorch, Graph Neural Networks scikit-learn, Supervised & RL C++, Python, R, SQL, SAS Gurobi, CPLEX, Xpress, Pyomo, Git

PROFESSIONAL EXPERIENCE

Graduate Researcher — Operations Research (MITACS Industrial Collaboration) — Rio Tinto, Montréal, QCJul. 2022 – Present

- Achieved a 90% reduction in mine-scheduling computation time via novel cutting planes and rolling-horizon heuristics deployed on Rio Tinto's industrial planning instances.
- Reduced time-to-feasibility by 80% through constraint relaxation strategies, enabling faster initial solution generation for large-scale MIP models.
- Translated Ph.D. research directly into production-relevant algorithmic tools, bridging academic OR and live mining operations.

Doctoral Researcher — Operations Research — Polytechnique Montréal · GERAD, Montréal, QC Aug. 2021 – Present

- Designed novel cutting planes for the Open-Pit Mine Production Scheduling Problem (OPMPSP), tightening LP relaxation bounds and reducing branch-and-bound search on 100,000+ variable instances.
- Developed relax-and-repair and parallel warm-start hybrid heuristic frameworks in C++ (OpenMP) for scalable feasible solution generation and improvement (second paper under review, Engineering Optimization).
- Currently building GNN and ML models (PyTorch, scikit-learn) to predict MILP instance difficulty and guide solver decisions, trained on a custom-generated dataset spanning six instance-generation strategies.
- Presented research at JOPT 2025 (Montréal), ISMP 2024 (Montréal), and IISE Annual Conference 2024.
- Author: "New cutting planes for open-pit mine scheduling." Computers & Operations Research, 2026.

Trainee Decision Scientist — Mu Sigma Inc., Bengaluru, India Sep. 2015 – Jul. 2016

- Delivered end-to-end analytics solutions for U.S. pharmaceutical and retail clients using SAS, SQL, R, and Tableau in an Agile environment.
- Built analytics pipelines, database management systems, and decision-support tools responding to ad-hoc client requests.

Research Intern — Flight Safety Analytics — École Nationale de l'Aviation Civile (ENAC), Toulouse, FranceJun. 2020 – Jan. 2021

- Developed a Python flight-safety analysis tool; applied data analysis and ML (random forest, regression) to Be58/TB20 and A340-300 flight datasets to identify unsafe operational conditions.

Graduate Research Assistant — Aerospace Materials — Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea Sep. 2016 – Jul. 2018

- Experimental and simulation-based research on aerospace composite structures (hypervelocity impact, morphing wing materials). Co-authored 3 peer-reviewed journals; 4 international conference presentations.

SELECTED PROJECTS ([GITHUB.COM/VENKAT-AKHIL-ANKEM](https://github.com/VENKAT-AKHIL-ANKEM))

- Mine Production AI Optimizer — GNN/ML pipeline for MILP difficulty prediction and solver guidance on open-pit scheduling instances.
- ML Scheduling Difficulty Predictor — supervised learning models (scikit-learn) predicting solve time and difficulty from instance structure.
- RL Job Scheduler — reinforcement learning approach to job/production scheduling problems.
- Research Paper AI Assistant — LLM-based tool supporting literature review and academic writing workflows.

EDUCATION

Ph.D. in Industrial Engineering (Operations Research) — Polytechnique Montréal · GERAD Aug. 2021 – Present (*expected 2027*)
Supervisors: Prof. Guy Desaulniers & Prof. Michel Gamache · Dissertation: Optimization methods for open-pit mine production planning · MITACS Fellow

M.Sc. in International Air Transport Operations Management — École Nationale de l'Aviation Civile (ENAC), Toulouse, France Aug. 2018 – Aug. 2021
OR in aviation: crew scheduling, route assignment, maintenance scheduling · GIFAS Scholar

M.Sc. in Aerospace Engineering — Korea Advanced Institute of Science and Technology (KAIST), South Korea Sep. 2016 – Jul. 2018
Aerospace composites & structures · KAIST Full Merit Scholarship

B.Tech. in Aerospace Engineering — Amrita Vishwa Vidyapeetham, Coimbatore, India Sep. 2011 – Jun. 2015
3rd in departmental ranking · Published novel solvent bonding technique (J. Adhesion Sci. Tech., 2015)

SELECTED PUBLICATIONS (5 PEER-REVIEWED JOURNAL ARTICLES)

OR: Ankem, V.A., Desaulniers, G., Gamache, M., & Raymond, V. "New cutting planes for open-pit mine scheduling with multi-period block extraction." *Computers & Operations Research*, 2026. doi.org/10.1016/j.cor.2026.107408

OR: Ankem, V.A., et al. "A relax-and-repair framework and a parallel warm-start hybrid heuristic for open-pit mine production scheduling." *Engineering Optimization (under review)*, 2026.

AE: Kim, Y., Ankem, V.A., et al. *Composite Structures*, 2018 · Sathish Kumar, S.K., Ankem, V.A., et al. *Composites Research*, 2018 · Nam, Y.-W., Ankem, V.A., et al. *Composite Structures*, 2018 · Ankem, V.A., et al. *J. Adhesion Sci. Tech.*, 2015.

Conferences: JOPT 2025 · ISMP 2024 · IISE 2024 · AACCM 2018 · KSCM 2017, 2018 · Advances in Composite Materials 2017 · ICTACEM 2014 (9 total)

GRANTS, FELLOWSHIPS & AWARDS

MITACS Fellowship — Ph.D. funding, Polytechnique Montréal & Rio Tinto · Sep. 2022

GIFAS Scholarship — ENAC, France · Sep. 2018

KAIST Full Merit Scholarship — graduate studies · Sep. 2016

Top 3 in B.Tech Aerospace Engineering Department — Amrita Vishwa Vidyapeetham · Mar. 2016

TECHNICAL SKILLS

Programming: C++, Python, R, SQL, SAS, C · Solvers: Gurobi, Xpress, Pyomo · HPC: OpenMP, Linux

Data & ML: PyTorch, Graph Neural Networks, NumPy, Pandas, scikit-learn, Tableau, Random Forest, Regression, Reinforcement Learning (applied) · Tools: Git, LaTeX, Agile

Languages: English (fluent), French (intermediate), Telugu (native), Hindi (fluent), Tamil (basic), Korean (basic).

LEADERSHIP & SERVICE

Student Representative & DEI Committee Member, GERAD (Sep. 2024 – Present) · Graduate Student Coordinator, ISSAC – KAIST (2017–18)

Director of Multimedia, KAIST International Students Association (2016–18) · ENAC Debate Club / MUN Lyon (2018–21)